2.6 THE SMITH CHART

The Smith Chart, shown in Figure 2.17, is a graphical tool for use with transmission line circuits and microwave circuit elements. It was created in the 1930s by Philip H. Smith, an engineer at Bell Telephone Labs, and was heavily used by engineers developing microwave systems in World War II. It was initially a very useful way of circumventing the complex-number arithmetic required in transmission line problems, only requiring a straight edge and a compass to operate. It has grown to be a very handy tool for understanding the behavior of microwave circuit elements and continues to be used, even though computers are quite capable of handling the complex math. In fact, it is common for Microwave CAD packages to display their solutions on a Smith Chart.

For hand calculations, the Smith Chart is most useful assuming the T-line is lossless. Although lossy lines can be modeled with the Smith Chart, it is not often practical to do so. Therefore, only lossless T-lines will be considered in the following discussion. The Smith Chart can be thought of as two graphs in one. First, it plots the normalized impedance at any point along a T-line. Second, it plots the reflection coefficient at any point along the line. Let's first look at how the chart is derived.

2.6.1 Smith Chart Derivation

As depicted in Figure 2.18(a), the complex reflection coefficient at a load is related to the load and line impedance as

$$\Gamma_L = \frac{Z_L - Z_o}{Z_L + Z_o} \tag{2.83}$$

We can normalize the load impedance to the characteristic impedance, writing

$$z_L = \frac{Z_L}{Z_o}$$

and then

$$\Gamma_L = \frac{z_L - 1}{z_L + 1} \tag{2.84}$$

Now, as Figure 2.18(b) suggests, we can replace the load along with any arbitrary length of T-line by an input impedance. The reflection coefficient at this new load is found from (2.66) to be

$$\Gamma = \Gamma_L e^{j2\beta z} = \frac{z - 1}{z + 1} \tag{2.85}$$

Here the reflection coefficient has the same magnitude as it does at the load but is phase shifted by $2\beta z$, where z is the distance along the line from the load to the input impedance point.

The reflection coefficient and normalized load terms are complex and can be expanded into real and imaginary parts as follows:

$$\Gamma = \Gamma_{Re} + j\Gamma_{Im} \tag{2.86}$$

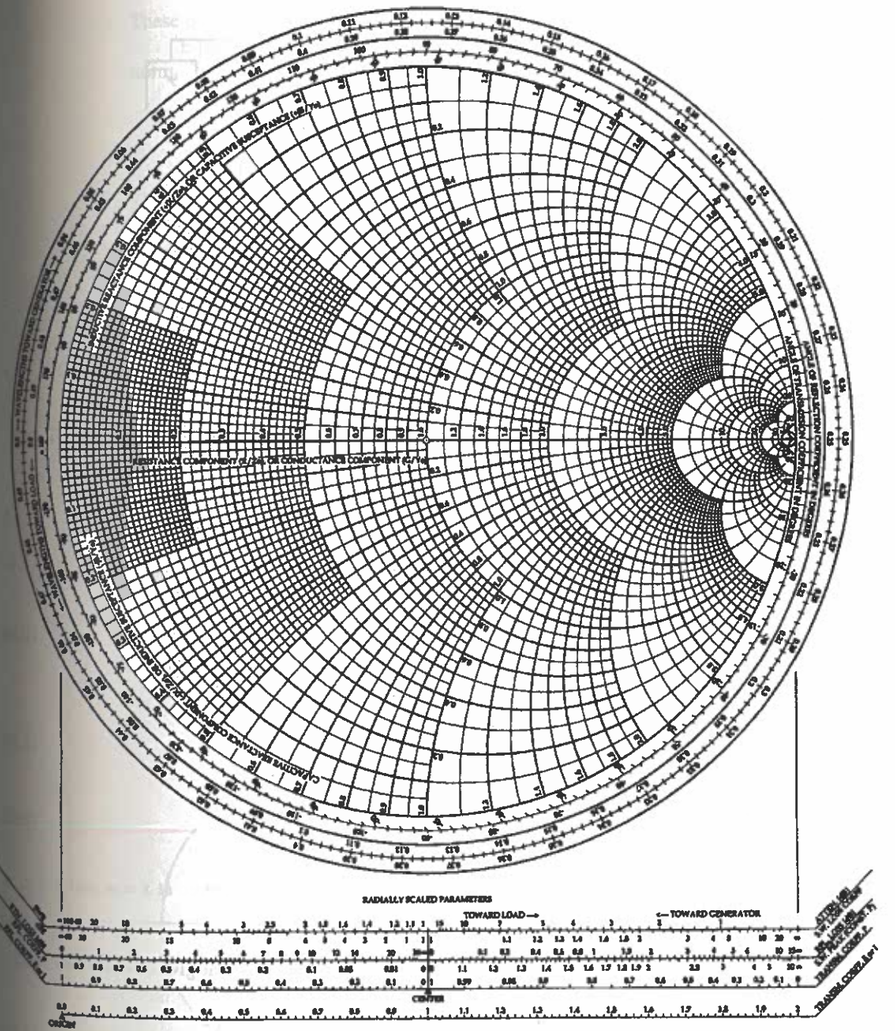


FIGURE 2.17 The *Transmission Line Calculator*, commonly referred to as the *Smith*® *Chart*.
®SMITH is a registered trademark of Analog Instrument Co., Box 950, New Providence, NJ 07974.
A printable version of the Smith chart along with other Smith chart resources is available at http://www.sss-mag.com/smith.html.

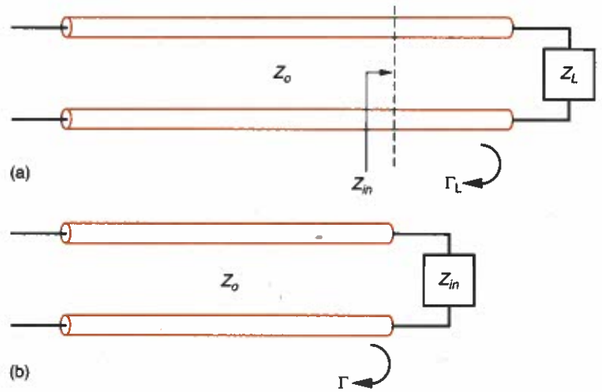


FIGURE 2.18 (a) T-line terminated in a load Z_L is characterized as having a reflection coefficient Γ_L. (b) A section of line containing the load is replaced by input impedance Z_{in}.

Rearranging (2.85) to solve for z in terms of Γ, we get

$$z = \frac{1+\Gamma}{1-\Gamma} \tag{2.87}$$

or

$$z = r + jx = \frac{1+\Gamma_{Re}+j\Gamma_{Im}}{1-\Gamma_{Re}-j\Gamma_{Im}} \tag{2.88}$$

Manipulating the right side of (2.88), we find the real part is

$$r = \frac{1-\Gamma_{Re}^2-\Gamma_{Im}^2}{(1-\Gamma_{Re})^2+\Gamma_{Im}^2} \tag{2.89}$$

and the imaginary part is

$$jx = \frac{j2\Gamma_{Im}}{(1-\Gamma_{Re})^2+\Gamma_{Im}^2} \tag{2.90}$$

Now, the general equation for a circle of radius a, centered at $x = m$ and $y = n$, is

$$(x-m)^2+(y-n)^2 = a^2 \tag{2.91}$$

Equations (2.89) and (2.90) can be rearranged into forms that give circular functions of r and x:

$$\left(\Gamma_{Re}-\frac{r}{r+1}\right)^2+\Gamma_{Im}^2 = \left(\frac{1}{r+1}\right)^2 \tag{2.92}$$

and

$$(\Gamma_{Re}-1)^2+\left(\Gamma_{Im}-\frac{1}{x}\right)^2 = \left(\frac{1}{x}\right)^2 \tag{2.93}$$

These circles will be plotted on the Γ_{Im} versus Γ_{Re} axes shown in Figure 2.19(a).

Equation (2.92) can be used to plot normalized *resistance circles*. Consider a normalized resistance $r = 1$. Then we have

$$\left(\Gamma_{Re}-\frac{1}{2}\right)^2+\Gamma_{Im}^2 = \frac{1}{4}$$

This is the equation for a circle centered at $\Gamma_{Re} = \frac{1}{2}$ and $\Gamma_{Im} = 0$ with a radius as shown in Figure 2.19(b). Also shown in this figure is the $r = 0$ circle, with a radius 1 centered at $\Gamma_{Re} = \Gamma_{Im} = 0$. The area within and on this $r = 0$ circle represents all the possible points for Γ, which must have a magnitude less than or equal to one. Similar circles can be drawn within this allowed space for any other value of r in the range $0 \le r \le \infty$.

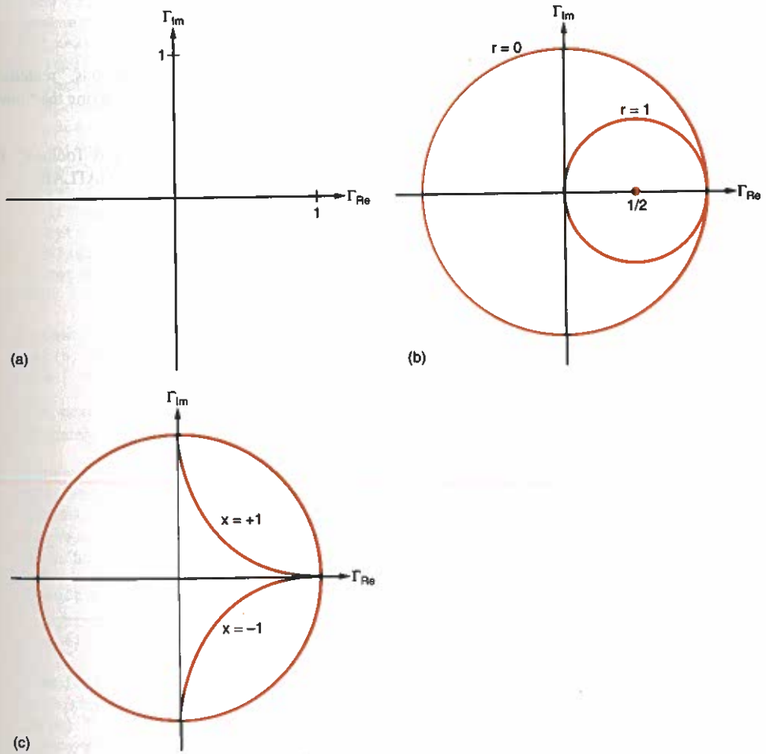


FIGURE 2.19 (a) Γ_{Re} versus Γ_{Im} axes for plotting the Smith Chart. (b) Circles of constant normalized resistance are added, followed by (c) circles of constant normalized reactance.

Using (2.93), we can plot normalized *reactance circles*. Consider a normalized reactance $x = 1$. Then we have

$$(\Gamma_{Re} - 1)^2 + (\Gamma_{Im} - 1)^2 = 1$$

This is a circle of radius 1 centered at $\Gamma_{Re} = \Gamma_{Im} = 1$. But notice that we only draw the circle for that part of the chart where $|\Gamma| \leq 1$. If the reactance is $x = -1$, we have

$$(\Gamma_{Re} - 1)^2 + (\Gamma_{Im} + 1)^2 = 1$$

or a radius 1 circle centered at $\Gamma_{Re} = 1$ and $\Gamma_{Im} = -1$. The allowed portions of these two circles are shown in Figure 2.19(c). More circles can be drawn for reactance, where it is seen that the upper half of the Smith Chart represents positive reactance (appearing inductive) and the bottom half represents negative reactance (appearing capacitive).

MATLAB 2.4

The following routine draws a Smith Chart. It calls a pair of functions, "realcirc" and "imcirc" to draw the lines. You can easily customize your own chart by modifying the "now add real circles" and "now add +/− x circles" portions of the program.

This routine was inspired by Soeren Laursen's "A Smith Chart Toolbox" from 1995, which contained a number of useful Smith Chart utilities written in MATLAB.

```
%   M-File: ML0204
%
%   This program plots a simple Smith Chart. It calls
%   on the functions realcirc, imcirc, and z2gamma.
%
%   Wentworth, 8/3/02
%
%   Variables:
%   a           circle radius
%   m,n         x,y center of circle
%   theta       angle (degrees)
%   z           a complex location
%   rvalues     values to plot for the real circles
%   xvalues     values to plot for the imag circles
%   xpos        x location of text
%   ypos        y location of text

clc             %clears the command window
clear           %clears variables

%first plot real = 0 circle
theta=linspace(-pi,pi,180);
a=1;
m=0; n=0;
Re=a*cos(theta)+m;
Im=a*sin(theta)+n;
z=Re+i*Im;
plot(z,'k')
```



```
axis('equal')
axis('off')
hold on

%add the x = 0 line
plot([-1 1],[0 0],'k');

%now add real circles
rvalues=[0.5 1 2 4];
for r=rvalues
   realcirc(r);
   xpos=z2gamma(r);
   h=text(xpos,0,num2str(r));
set(h,'VerticalAlignment','top','HorizontalAlignment','right');
end

%now add +/-x circles
xvalues=[0.2 0.5 1 2];
for x=xvalues
   imcirc(x);
   imcirc(-x);
   xpos=real(z2gamma(i*x));
   ypos=imag(z2gamma(i*x));
   h=text([xpos xpos],[ypos -ypos],[' j' num2str(x);'-j' num2str(x)]);
   set(h(1),'VerticalAlignment','bottom');
   set(h(2),'VerticalAlignment','top');
   if xpos==0
   set(h,'HorizontalAlignment','center');
   elseif xpos<0
   set(h,'HorizontalAlignment','right');
end

h=text(-1,0,'0');
set(h,'VerticalAlignment','middle','HorizontalAlignment','right');
end
```

The next three functions (z2gamma(r), realcirc(r), and imcirc(r)) were entered and saved separately.

```
function[result] = z2gamma(z)
%Z2GAMMA Convert impedance to reflection coefficient
%GAMMA = Z2GAMMA(Z)
%converts the impedances in matrix Z to reflection
%coefficients in GAMMA.

result = (z-1)./(z+1);
```

```
function[h]=realcirc(r)

%REALCIRC(r) draws circle of constant real with
%normalized r;
phi=1:1:360;
theta=phi*pi/180;
a=1/(1+r);
m=r/(r+1);n=0;
```

```
Re=a*cos(theta)+m;
Im=a*sin(theta)+n;
z=Re+i*Im;
h=plot(z,'k');
axis('equal')
axis('off')
```

```
function[h]=imcirc(x)
%IMCIRC(x) draws a circle of constant imag with
%normalized x;
a=abs(1/x);
m=1;
n=1/x;
k=1;
for t=1:1:360
   angle(t)=t*pi/180;
   Re(t)=a*cos(angle(t))+m;
   Im(t)=a*sin(angle(t))+n;
   z(t)=Re(t)+i*Im(t);
   if abs(z(t)) <= 1
            zz(k)=z(t);
             k=k+1;
   end
end
h=plot(zz,'k');
axis('equal')
axis('off')
```

A Smith Chart generated using this routine is displayed in Figure 2.20.

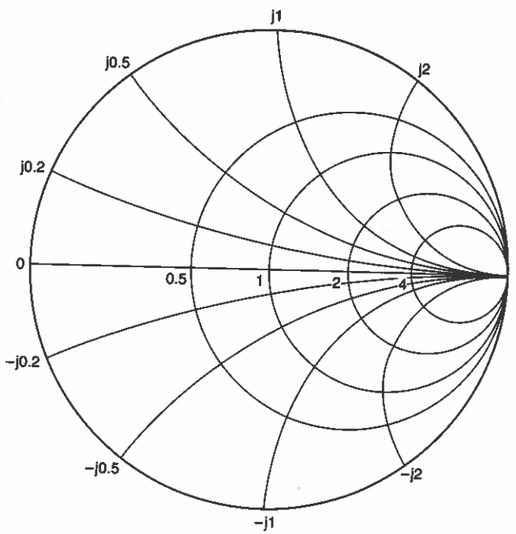
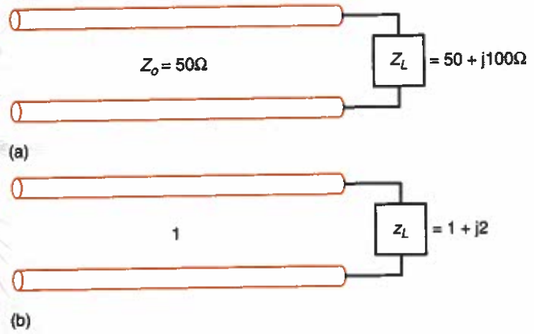


FIGURE 2.20 Smith Chart generated using ML0204.

2.6.2 Using the Smith Chart

The Smith Chart is a plot of normalized impedance. For instance, suppose you have a $Z_o = 50\ \Omega$ line terminated in a load $Z_L = 50 + j100\ \Omega$ as shown in Figure 2.21(a). To locate this point on the Smith Chart, you would first normalize the load impedance to obtain $z_L = 1 + j2$



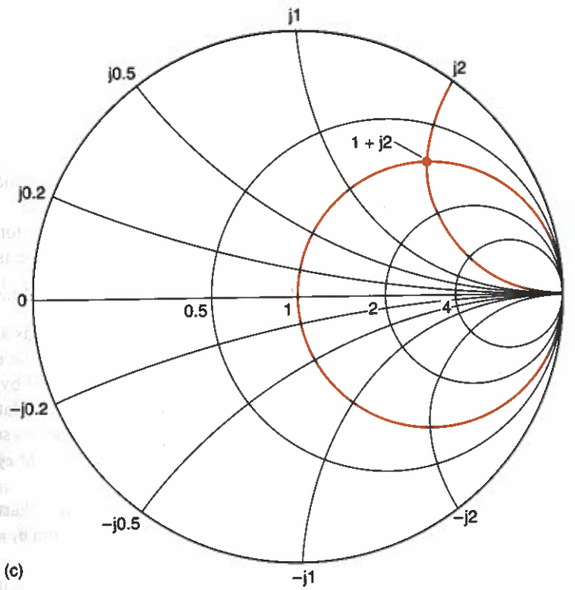


FIGURE 2.21a A T-line terminated in a load (a) shown with values normalized to Z_o in (b). (c) The location of the normalized load impedance is found on the Smith Chart.

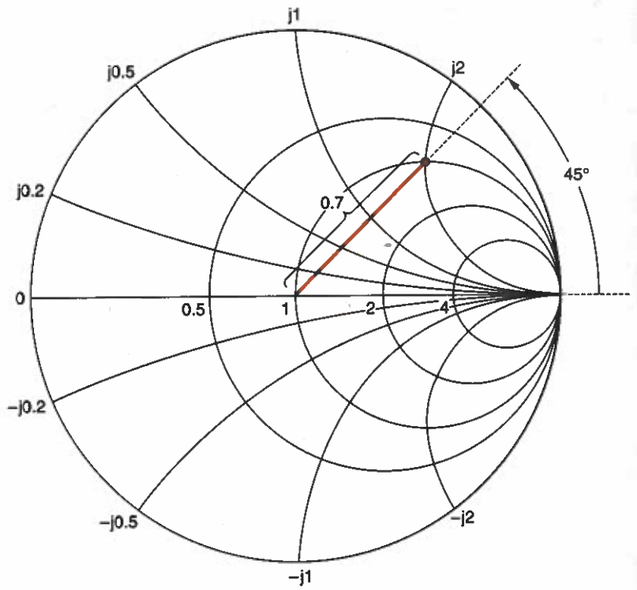


FIGURE 2.21b (d) Reflection coefficient and angle.

(Figure 2.21(b)). Then, as shown in Figure 2.21(c), the normalized impedance is located at the intersection of the $r = 1$ circle and the $x = +2$ circle.

To see how the Smith Chart is also a plot of the reflection coefficient, we recall that the reflection coefficient at any point z along a lossless T-line is

$$\Gamma = \Gamma_{Re} + j\Gamma_{Im} = \Gamma_L e^{j2\beta z} = |\Gamma_L| e^{j\theta_\Gamma} \tag{2.94}$$

In this equation, we note that the reflection coefficient has a magnitude $|\Gamma_L|$ and an angle θ_Γ equal to its angle at the load plus $2\beta z$. The actual value of $|\Gamma_L|$ is found by taking the distance from the center of the chart to the point divided by the distance from the center of the chart to the periphery ($|\Gamma_L| = 1$). To avoid this calculation, a scale for magnitude of reflection coefficient is provided below the Smith Chart, as seen in Figure 2.17. The angle of the reflection coefficient θ_Γ is indicated on the *angle of reflection coefficient* scale, shown just outside the $|\Gamma_L| = 1$ circle on the chart.

Continuing our example, we read from the Smith Chart (see Figure 2.21(d)) that the normalized load $z_L = 1 + j2$ corresponds to $|\Gamma_L| = 0.7$ and $\theta_\Gamma = 45°$. This can be verified by calculation:

$$\Gamma_L = \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{50 + j100 - 50}{50 + j100 + 50} = 0.5 + j0.5 = 0.707 e^{j45°}$$

The values of normalized impedance and reflection coefficient are functions of position along the T-line (see Figure 2.22). After locating the normalized impedance point, it is useful to draw the *constant-$|\Gamma_L|$ circle*. Since

$$\Gamma = |\Gamma_L| e^{j\theta_\Gamma}$$

we let θ_Γ change as we hold $|\Gamma_L|$ constant, and this traces out a circle of constant $|\Gamma_L|$ on the chart. Now, moving along this constant-$|\Gamma_L|$ circle is akin to moving along the T-line. Recalling that $\theta_\Gamma = 2\beta z + \phi$, we see that the input impedance at some point $z = -\ell$ along the T-line corresponds to $\theta_\Gamma = -2\beta\ell + \phi$. Thus, moving away from the load corresponds to moving in the clockwise direction on the Smith Chart.

Since the function $e^{j\theta_\Gamma}$ is sinusoidal, it repeats for

$$2\beta z = N2\pi$$

(where $N = 1, 2, 3\ldots$). So, since $\beta = 2\pi/\lambda$, we see that at values of z where

$$z = N\frac{\lambda}{2}$$

Γ must repeat, and this distance corresponds to travel of an integral multiple of half wavelengths along the line.

Continuing our example, we suppose the line is 0.3λ in length. Drawing a line from the center of the chart to the outside *Wavelengths Toward Generator* (WTG) scale,[2.6] as shown in Figure 2.22(b), we observe a starting point at 0.188λ. Adding 0.3λ moves us along the constant-$|\Gamma_L|$ circle to 0.488λ on the WTG scale, corresponding to a normalized input impedance of $z_{in} = 0.175 - j0.08$. Denormalizing, we find an input impedance of

$$Z_{in} = z_{in} Z_o = 8.75 - j4\,\Omega$$

The voltage standing wave ratio can be determined by reading the value of r at the $\theta_\Gamma = 0°$ crossing for the constant-$|\Gamma_L|$ circle (point b in Figure 2.20). This can be seen by first recalling that

$$z = r + jx = \frac{1 + \Gamma}{1 - \Gamma}$$

Notice that, at $x = 0$ and $r > 1$, $|\Gamma_L| = \Gamma$, so

$$z = r = \frac{1 + |\Gamma|}{1 - |\Gamma|} = VSWR$$

In the continuing example, we find $VSWR = 5.9$ by reading from the Smith Chart. More exact calculation gives 5.83.

The point where *VSWR* is taken is also the location of the maximum value of r along the constant-$|\Gamma_L|$ circle. The minimum value of r occurs a distance $\lambda/4$ away (at $\theta_\Gamma = 180°$). These two points for maximum and minimum values of r correspond to the locations of relative voltage maximum and minimum on the T-line.

[2.6] The WTG scale is used when moving clockwise, or away from the load. The *Wavelengths Toward Load* (WTL) scale is used when moving counterclockwise on the chart.

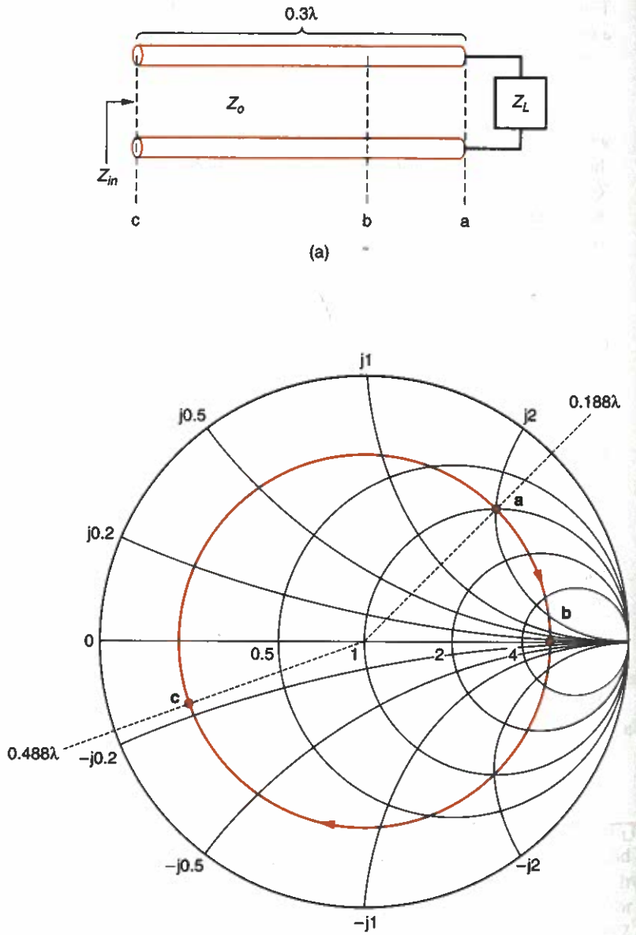


FIGURE 2.22 Movement along a T-line corresponds to movement along a constant-$|\Gamma_L|$ circle on the Smith Chart.

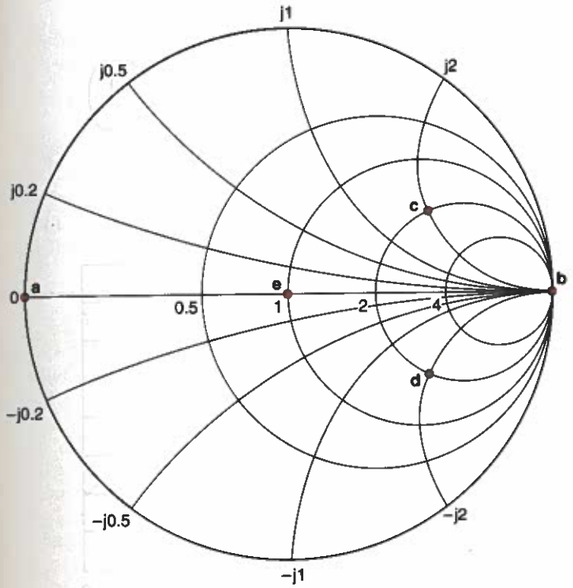


FIGURE 2.23 Solution for Drill 2.11.

Drill 2.11 Locate the following load impedances terminating a 50-Ω T-line: (a) $Z_L = 0$ (a short circuit), (b) $Z_L = \infty$ (an open circuit), (c) $Z_L = 100 + j100\ \Omega$, (d) $Z_L = 100 - j100\ \Omega$, and (e) $Z_L = 50\ \Omega$. (*Answer:* See Figure 2.23)

Drill 2.12 A 0.334λ-long $Z_o = 50$-Ω T-line is terminated in a load $Z_L = 100 - j100\ \Omega$. Use the Smith Chart to find (a) Γ_L, (b) *VSWR*, (c) Z_{in}, and (d) the distance from the load to the first voltage minimum. (*Answer:* (a) $0.61e^{-j30^\circ}$, (b) 4.3, (c) $Z_{in} = 22.5 + j45\ \Omega$, and (d) 0.208λ)

2.6.3 Impedance Measurement

Measuring a device's input impedance requires connecting it to a measuring instrument. However, the presence of the connection cable between the device and the measurement apparatus can significantly influence the measured impedance, especially at high frequency. Hence, several schemes for accurately measuring impedance have been devised.

One technique for measuring input impedance employs a slotted coaxial air line, as shown in Figure 2.24(a). A probe can be slid along the slotted line to measure electric field amplitude. A scale is affixed to the line for accurate location of the voltage maxima and minima. The ratio of the voltage maxima to the minima is the *VSWR*. To determine

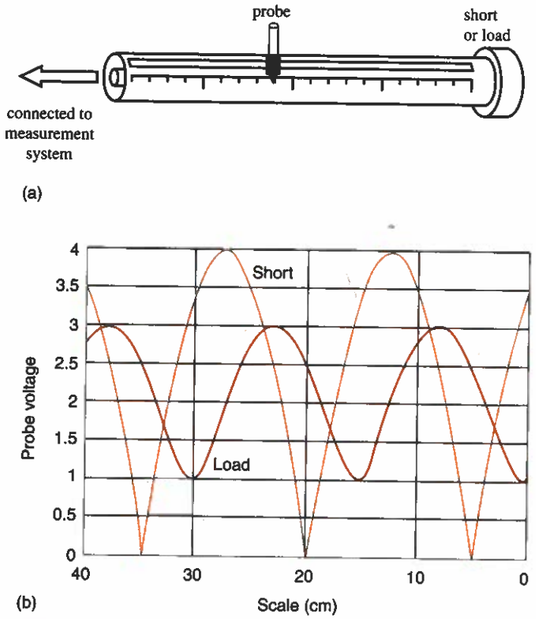


FIGURE 2.24a (a) Slotted coaxial air line. (b) Field magnitude versus position along the slotted line.

the input impedance, the field magnitude is plotted with the load in place and compared to the field plotted with the load replaced with a short circuit. An example details this procedure.

EXAMPLE 2.4

An unknown load is attached to a 50-Ω impedance slotted coaxial air line. The load is determined with the following procedure:

1. Terminate the line in a short circuit. Determine the guide wavelength and the location of the voltage minima. Choose one of these minima as the reference location of the load.

   The light blue line in Figure 2.24(b) shows the field amplitude plotted against the scale position for a short-circuit termination. The "0 cm" end of the scale is close to the load, but not necessarily right at the load. Voltage minima are located at 5, 20, and 35 cm, and we'll arbitrarily choose 5 cm to be the reference location for our load. From our understanding of the Smith Chart we know the difference in two minima (15 cm) corresponds to λ/2. So λ = 30 cm, and because we are dealing with an air line ($u_p = c$) we have $f = c/\lambda = 1$ GHz.

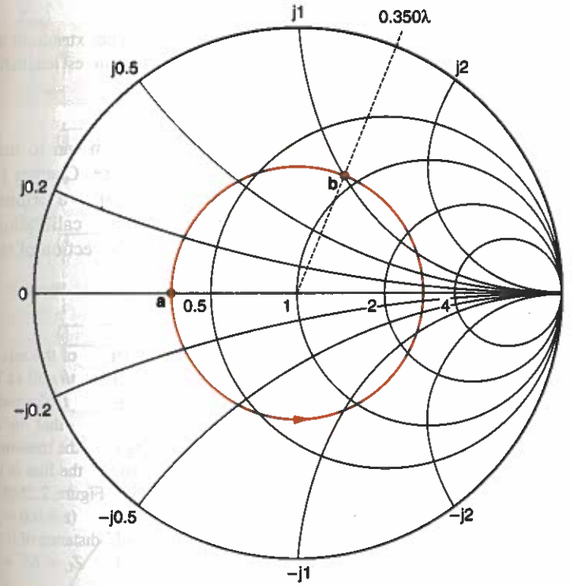


FIGURE 2.24b (c) Smith Chart solution to Example 2.4.

2. Terminate the line in the load. Determine the *VSWR* and draw a constant-|Γ|circle. Also determine the location of voltage minima.

   The heavy blue line in Figure 2.24(b) represents the field plot with the load in place that has a voltage maxima of 3 and a minima of 1, so we have a *VSWR* = 3. This is used to draw the constant-|Γ| circle on the Smith Chart of Figure 2.24(c). Finally, from Figure 2.24(b), we see that the voltage minima with the load in place are located at 0.5, 15.5, and 30.5 cm.

3. Move from one of the load minima to the reference location of the load.

   On the Smith Chart, we can begin at the 15.5-cm voltage minimum (point *a*). If we move to our 5-cm reference location, then we must move 10.5 cm or

$$\frac{10.5\,\text{cm}}{30\,\text{cm}/\lambda} = 0.350\lambda$$

   toward the load. This is point *b* on our constant-|Γ| circle, corresponding to $z = 0.8 + j1.0$, or

$$z_L = 40 + j50\,\Omega$$

   Had we instead begun at the 0.5-cm minimum, then we would have moved 4.5 cm (0.150λ) toward the generator and reached the same point b. ■

Drill 2.13 Suppose in Example 2.4 that the 50-Ω coaxial air line extends all the way from the 0-cm scale location to the location of the termination. What is the shortest length this extension can be? (*Answer:* 10 cm)

Rather than using a slotted line, it is much more common to measure the input impedance of a device or network using a *network analyzer* (see Chapter 10). The effect of the connection cable can be *calibrated out* by using a set of standard terminations. To get an idea of how this is accomplished, the following example uses a calibration short circuit to determine the impedance of a load connected to the end of a section of transmission line.

EXAMPLE 2.5

The calibration short circuit is used to determine the electrical length of the test connection to the location of the device to be tested. Suppose when we attach a short circuit to a 50-Ω T-line, as indicated in Figure 2.25(a), that we measure an input impedance of $Z_{insc} = +j128\ \Omega$. Normalizing this to 50 Ω, we find $z = j2.56$, or point a on the chart of Figure 2.25(c). We know that the actual normalized impedance of a short is $z = 0 + j0$ (point b on the chart). Moving from the measured short at point a toward the actual load at point b, we see that the electrical length of the line is 0.191λ.

Now suppose we attach the unknown load, as indicated in Figure 2.25(b), and measure an impedance of $Z_{inL} = 30 - j40\ \Omega$. We normalize and locate this as point c ($z = 0.6 - j0.8$). This point is located at 0.125λ on the WTL scale. We then move toward the load a distance of 0.191λ, or to 0.316λ on the WTL scale (point d). This corresponds to $z_L = 1.3 + j1.4$, or $Z_L = 65 + j65\ \Omega$. ■

The Smith Chart is used in a variety of microwave engineering applications, in particular impedance matching and amplifier design. Impedance matching is the topic of the next section, and it will be revisited in Chapter 10 prior to the topic of amplifier design.

2.7 IMPEDANCE MATCHING

We often desire that all of the power propagating along a T-line be dropped across the terminating load impedance. But as we've seen, an impedance mismatch can result in the reflection of much of this power. An impedance *matching network*, as indicated in Figure 2.26, provides a solution. The impedance looking into the network is matched to the line impedance. If the network itself consists only of reactive elements, then it will dissipate no power, and consequently all the power will be dropped across the load.

A number of techniques are available for constructing matching networks. Practicality is one concern, since the network should be of fairly simple design and be easy to implement. The most practical designs include quarter-wave transformers, single-stub tuners, lumped-element tuners, and multisection transformers. With the exception of multisection transformers, all of these matching networks tend to operate over a fairly narrow bandwidth. One desirable trait for these networks is to have tuning capability in the event that the load impedance changes. Using variable reactive elements in a lumped-element tuner or using an adjustable-length T-line stub in a stub tuner can provide some tuning capability.

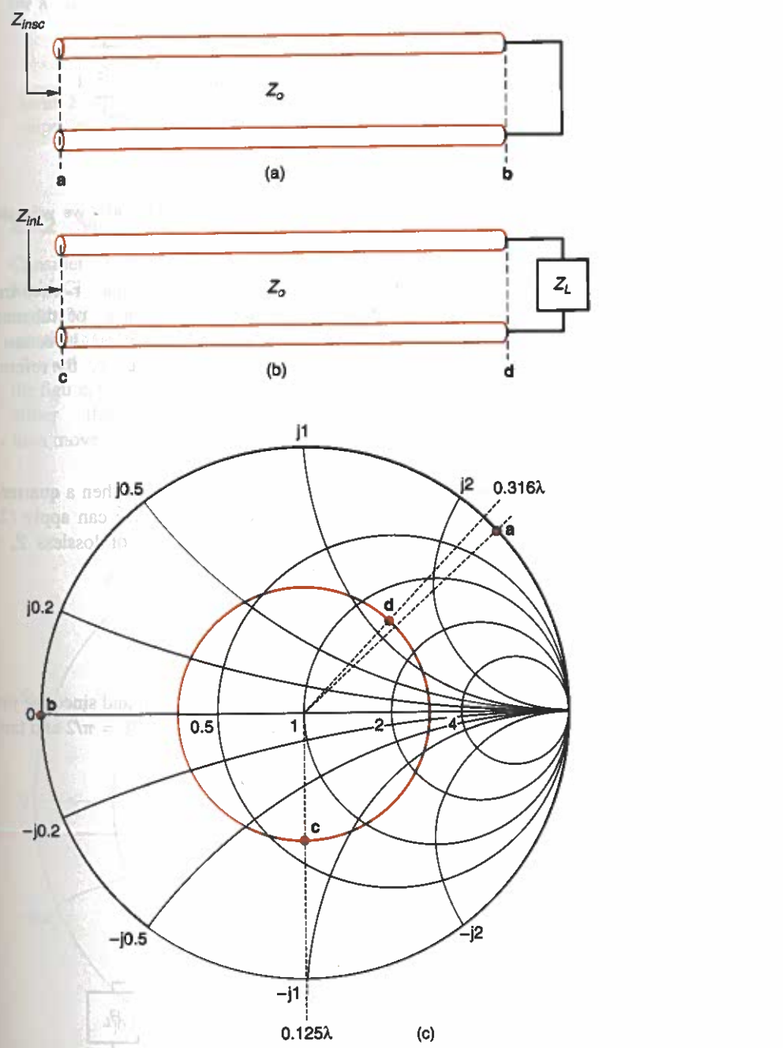


FIGURE 2.25 The short-circuited T-line of (a) determines the electrical length of the line that is then used in (b) to find the unknown load with the aid of a Smith Chart (c).

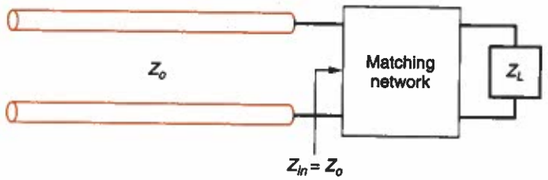


FIGURE 2.26 Adding an impedance-matching network ensures that all power will make it to the load.

In this section we will first investigate the simple quarter-wave transformer. Then we'll see how the Smith Chart may be used to construct a variety of stub matching networks. Lumped-element networks will be deferred until Chapter 10. Multisection tuning networks are beyond the scope of this text but can be studied in several of the references cited at the end of Chapter 10.

2.7.1 Quarter-Wave Transformer

If the load impedance is all real (no reactive component), then a quarter-wave matching network can be constructed as shown in Figure 2.27. We can apply (2.73) to find the impedance looking into the quarter-wave-long section of lossless Z_s impedance line terminated in a resistive load R_L. We have

$$Z_{in} = Z_S \frac{R_L + jZ_S \tan \beta\ell}{Z_S + jR_L \tan \beta\ell}$$

The line length is a quarter of a wavelength, or $\ell = \lambda_g/4$, and since the phase constant on the line is related to wavelength as $\beta = 2\pi/\lambda_g$, we have $\beta\ell = \pi/2$ and $\tan(\beta\ell) = \infty$. This leads to

$$Z_{in} = \frac{Z_S^2}{R_L} = Z_o$$

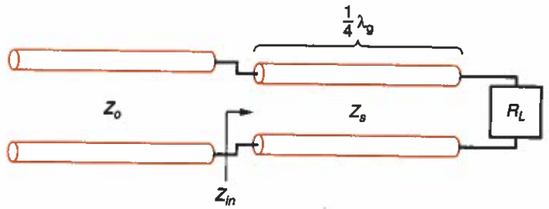


FIGURE 2.27 Quarter-wave transformer.

for an impedance-matched line, or

$$Z_S = \sqrt{Z_o R_L} \tag{2.95}$$

Drill 2.14 Suppose a 50-Ω T-line is terminated in a 100-Ω load. Determine the required impedance of a quarter-wave matching section of T-line. (*Answer:* 70.7 Ω)

2.7.2 Matching with the Smith Chart

Consider a normalized load impedance z_L located at an arbitrarily selected point on the Smith Chart of Figure 2.28. Since, as shown in Figure 2.26, we want $Z_{in} = Z_o$, the normalized input impedance will be located at the center of the Smith Chart, where $|\Gamma| = 0$. Our matching network must therefore move us from z_L to the center of the chart.

Moving away from the load along the Z_o line generates the constant-$|\Gamma|$ circle shown in the figure. Of special interest are the two points where this circle crosses the $1 \pm jx$ circle. At either of these points, inserting the appropriate reactive element (that is, adding $\mp jx$) will then move us to the matched condition at the center of the chart.

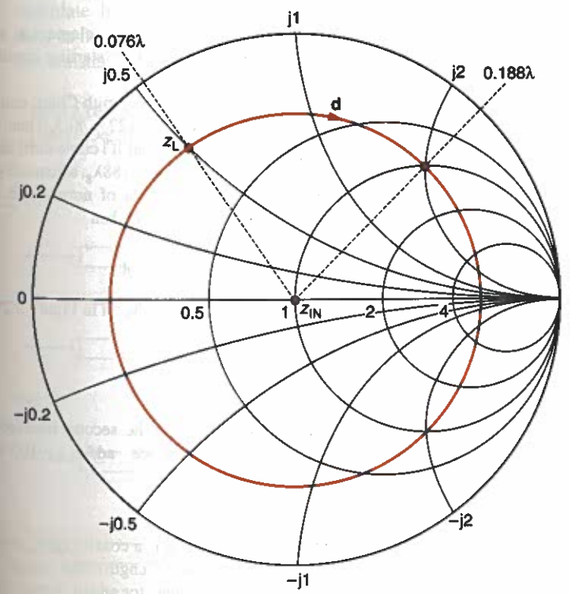


FIGURE 2.28 The objective of a matching network is to move to the center of the Smith Chart. The values shown are for Example 2.6.

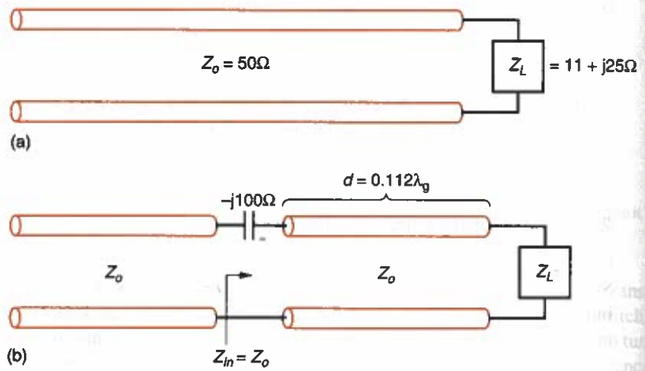


FIGURE 2.29 (a) A 50-Ω T-line terminated in an $11 + j25\ \Omega$ load for Example 2.6. (b) T-line with tuning capacitor added at the appropriate distance from the load.

EXAMPLE 2.6

Let us construct a simple matching network by adding a reactive element at a suitable location along a 50-Ω T-line terminated in an $11 + j25\ \Omega$ load. This starting condition is shown in Figure 2.29(a).

We first locate the normalized load impedance z_L on the Smith Chart, and then we draw the constant-$|\Gamma|$ circle. This is shown in Figure 2.28, where $z_L = 0.22 + j0.5$. Then, we move from z_L clockwise (toward the generator) a distance d along the constant-$|\Gamma|$ circle until the point $1 + j2.0$ is reached. Using the WTG scale, we are moving from $0.076\lambda_g$ to $0.188\lambda_g$, a distance of $0.112\lambda_g$. We cut the T-line at this point and insert a series capacitive element of normalized reactance $-j2.0$, corresponding to $-j100\ \Omega$. The normalized input impedance is then

$$z_{in} = 1 + j2.0 - j2.0 = 1 + j0$$

corresponding to the center of the Smith Chart. The result is shown in Figure 2.29(b). The value of capacitance required depends on frequency, that is,

$$-j100\ \Omega = \frac{-j}{\omega C}$$

Note that we could have proceeded along Z_o until the second intersection was reached (at $1 - j2.0$), in which case a series inductor could have been added ($+j100\ \Omega$) to provide the impedance match. ■

Drill 2.15 Suppose, for Example 2.6, that the 50-Ω line is a coaxial cable made with a Teflon dielectric, and it must operate at 800 MHz. Determine (a) the length of the coaxial line between the load and the capacitor and (b) the value of the series capacitor added to provide an impedance match. (*Answer*: $d = 2.9$ cm, (b) $C = 2.0$ pF)

Before going on to shunt-stub matching, we first need to discuss the admittance of open-ended and shorted T-line stubs.

2.7.3 Admittance of Shunt Stubs

Sometimes, as in the case of T-line stubs, we find it much more convenient to add shunt elements rather than series elements. With shunt elements, it is much easier to work in terms of admittances. Admittance is the inverse of impedance, allowing direct addition of parallel elements. Figure 2.30(a) shows the relationship between admittances and impedances. Here we see that the characteristic admittance Y_o is simply $1/Z_o$, and the load admittance Y_L is $1/Z_L$. The convenience of admittances is that shunt values may be added, for instance as shown in Figure 2.30(b).

It may be noted that the Smith Chart is also a chart of normalized admittance. The normalized load admittance is

$$y_L = \frac{Y_L}{Y_o} = \frac{1}{z_L} \tag{2.96}$$

It is customary to represent the normalized admittance as $y = g + jb$.

Consider the normalized impedance $z_L = 2 + j1$ as shown in Figure 2.31. We can calculate the normalized admittance as $y_L = 1/z_L = 0.40 - j0.20$. With the Smith Chart it is easy to find the normalized admittance. We simply move to a point on the opposite side of the constant-$|\Gamma|$ circle, as depicted in the figure.

Drill 2.16 A 50-Ω line is terminated in a pair of parallel load impedances of $50 + j100\ \Omega$ and $50 - j100\ \Omega$. Determine the total load admittance and impedance seen by the line. (*Answer*: 8 mS, 125 Ω)

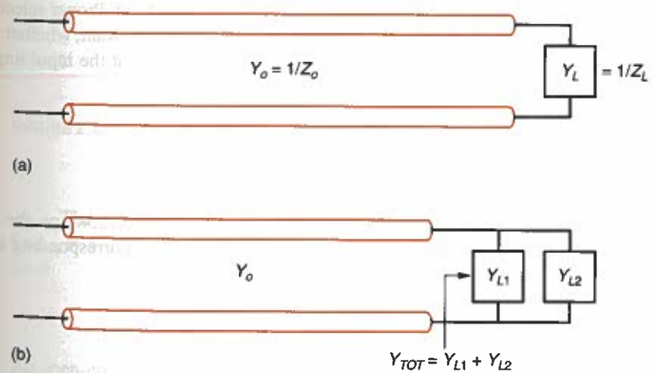


FIGURE 2.30 (a) Admittance relationship to impedance. (b) Adding shunt elements using admittances.

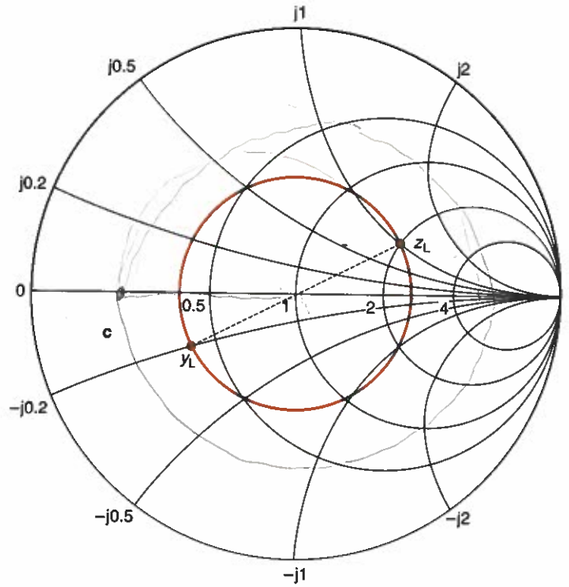


FIGURE 2.31 Finding normalized admittance with a Smith Chart.

As pointed out in the special terminations portion of Section 2.4, lossless T-line stubs that are terminated in either a short or an open end can be used as purely reactive tuning elements. Consider a shorted T-line as shown in Figure 2.32(a). We see that the constant-|Γ| circle follows the periphery of the Smith Chart ($z = 0 \pm jx$). Proper selection of the T-line length d allows us to choose any value of reactance that we want, whether it's capacitive or inductive. This is verified from (2.73), where it is seen that the input impedance at some distance d from the short is

$$Z_{in} = jZ_o \tan(\beta d) \tag{2.97}$$

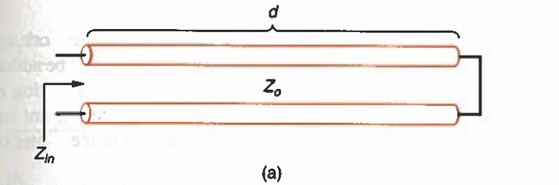
or

$$z_{in} = j \tan(\beta d) \tag{2.98}$$

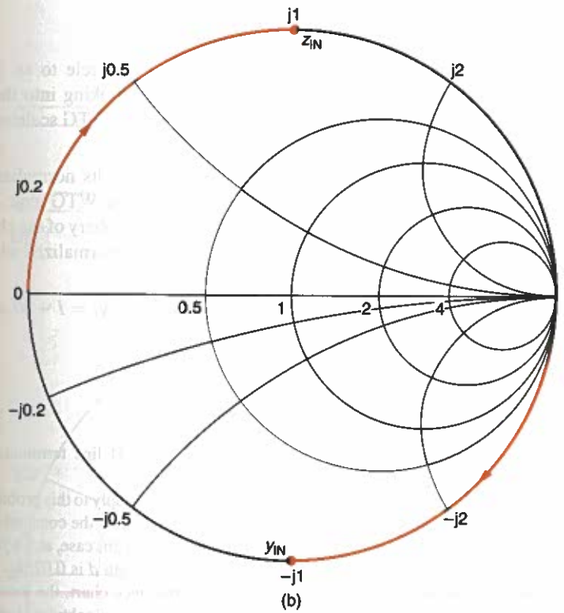
For instance, if we travel from the short to a distance $0.125\lambda_g$ on the WTG scale, the Smith Chart gives us an input impedance $Z_{in} = +jZ_0\ \Omega$, corresponding to an inductance. This is confirmed by using (2.97), where

$$\beta d = \left(\frac{2\pi}{\lambda_g}\right)\left(\frac{\lambda_g}{8}\right) = \frac{\pi}{4}$$

results in $\tan(\pi/4) = 1.0$, leading to $Z_{in} = jZ_0\ \Omega$.



(a)



(b)

FIGURE 2.32 (a) A shorted T-line stub. (b) Smith Chart view of z_{in} and y_{in} for $d = \lambda/8$.

The stub admittance is found by starting from the point on the right side of the chart ($y_{short} = \infty + j\infty$), and traveling clockwise to the point $y_{in} = 0 - j1.0$. Mathematically, we have

$$y_{in} = -j \cot(\beta d) \tag{2.99}$$

An open-ended stub will be very similar. In Figure 2.32(b), the z_{in} and y_{in} curves would be interchanged for an open stub.

2.7.4 Shunt-Stub Matching

In Smith Chart terms, the objective of a shunt-stub matching network as shown in Figure 2.33(a) is to move to the center of the chart. Since a shunt stub will be added, we will work in the admittance chart. From the normalized load admittance, a section of through line is traversed to arrive at the $1 \pm jb$ circle. At this point we add a shunt stub of normalized admittance $0 \mp jb$. The sum of these admittances takes us to the center of the chart where $|\Gamma| = 0$, $Y_{in} = Y_o$, and $Z_{in} = Z_o$, and matching is complete.

The procedure for constructing a stub matching network is as follows:

1. Locate z_L, the normalized load impedance.
2. Draw the constant-$|\Gamma|$ circle and use it to locate y_L.
3. From y_L, move clockwise along the constant-$|\Gamma|$ circle to an intersection with the $1 \pm jb$ circle, at which point the impedance looking into the through line is $y_d = 1 \pm jb$. The distance moved is found using the WTG scale and represents the through-line length d.
4. If a shorted shunt stub is employed, consider that its normalized admittance is located on the periphery of the chart at $0.250\lambda_g$ on the WTG scale, or at $\infty + j\infty$ on the admittance chart. Move clockwise along the periphery of the chart to $0 \mp jb$. The distance traveled is the length of the stub, ℓ, and the normalized admittance looking into the stub is therefore $y_\ell = \mp jb$.
5. The total admittance (Figure 2.33(b)) is $y_{tot} = y_d + y_\ell = 1 + j0$, and the matching network is complete.

EXAMPLE 2.7

Construct the shorted shunt-stub matching network for a 50-Ω line terminated in a load $Z_L = 20 - j55\ \Omega$.

The points indicated in the Smith Chart of Figure 2.33(c) apply to this problem. First, we locate the normalized load impedance, $z_L = Z_L/Z_o = 0.4 - j1.1$, and draw the constant-$|\Gamma|$ circle. Then we locate y_L. Moving to the first intersection with the $1 \pm jb$ circle (in this case, at $1 + j2.0$), we travel from $0.112\lambda_g$ to $0.187\lambda_g$ on the WTG scale, so our through-line length d is $0.075\lambda_g$.

Next we insert the shorted shunt stub. On the admittance chart, the location of the short is on the right side of the chart at WTG $= 0.250\lambda_g$. We must move clockwise (toward the generator) until we reach the point $0 - j2.0$, located at WTG $= 0.324\lambda_g$. This gives us a stub length of $0.324\lambda_g - 0.250\lambda_g = 0.074\lambda_g$. ■

EXAMPLE 2.8

Now we want to construct an open-ended shunt-stub matching network for a 50-Ω line terminated in a load $Z_L = 150 + j100\ \Omega$.

Referring to Figure 2.34(b), we first locate $z_L = Z_L/Z_o = 3.0 + j2.0$ and draw the constant-$|\Gamma|$ circle. Then we locate y_L. Moving to the first intersection with the $1 \pm jb$ circle (in this case, at $1 + j1.6$), we travel from WTG $= 0.474\lambda_g$ to WTG $= 0.178\lambda_g$. We add a half a wavelength to the end point, so the length of our through line is $d = (0.500\lambda_g) + 0.178\lambda_g - 0.474\lambda_g = 0.204\lambda_g$.

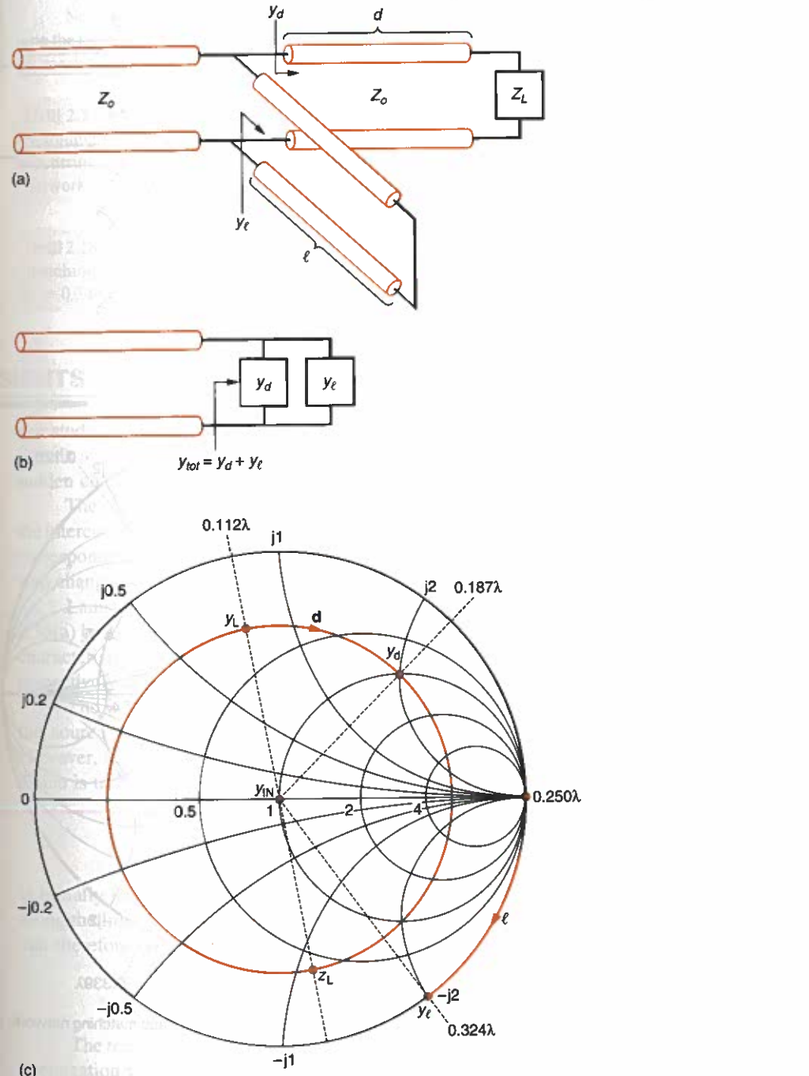
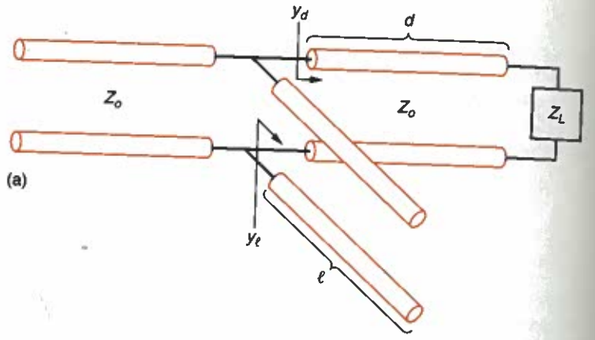


FIGURE 2.33 (a) The generic layout of the shorted shunt-stub matching network. (b) Adding shunt admittances. (c) Using the Smith Chart to find through line and stub lengths. Values on the chart apply to Example 2.7.



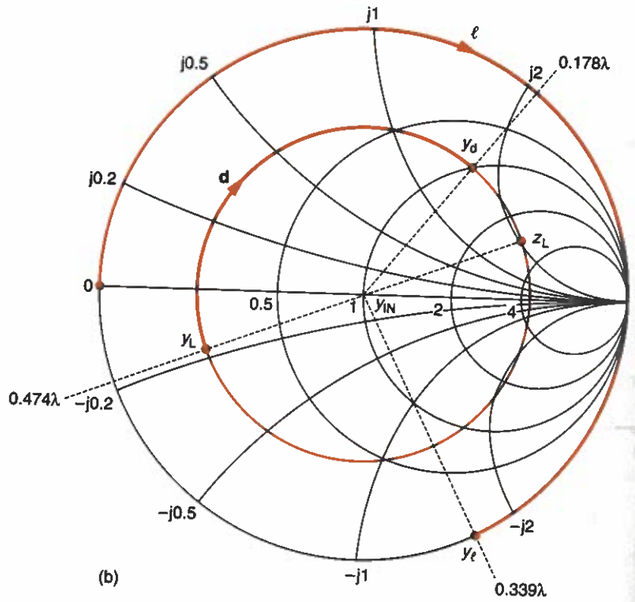


FIGURE 2.34 (a) The generic layout of the open-ended shunt-stub matching network. (b) Smith Chart solution to Example 2.8.

Next we insert the open-ended shunt stub. On the admittance chart, the location of the open end is on the left side of the chart at WTG = $0.000\lambda_g$. We must move clockwise (toward the generator) until we reach the point $0 - j1.6$ located at WTG = $0.339\lambda_g$, for a stub length $\ell = 0.339\lambda_g$. ■

Drill 2.17 In Example 2.7, we chose the first intersection with the $1 \pm jb$ circle (at $1 + j2.0$) in designing our matching network. We could also have continued on to the second intersection, occurring at $1 - j2.0$. Determine the through-line length d, and the stub length ℓ, for the matching network using this second intersection. (*Answer:* $d = 0.201\lambda_g$, $\ell = 0.426\lambda_g$)

Drill 2.18 Determine the through-line length d and the stub length ℓ for the open-ended shunt stub matching network of Example 2.8 if the other intersection with the $1 \pm jb$ circle is used. (*Answer:* $d = 0.348\lambda_g$, $\ell = 0.161\lambda_g$)

2.8 TRANSIENTS

Our study of transmission lines has so far concerned propagation of steady-state sinusoidal signals at a single frequency. We are now interested in the *transient* situation, where a sudden change in voltage or current is instigated at one end of a line.

The most common example of transients on T-lines is the propagation of signals along the interconnects between digital circuits. Information is carried as ones and zeros, typically corresponding to voltage levels of 6 and 0 V. Switching from 0 to 6 V, for instance, involves a step change in voltage that propagates along the interconnect.

Launching a voltage step function on a T-line can be accomplished as shown in Figure 2.35(a) by closing the switch at $t = 0$. For simplicity, we assume this T-line is lossless with characteristic resistance R_o. We'll also assume resistances R_s and R_L at the source and load, respectively. Reactive terminations will be dealt with later in this section.

The voltage launched at the T-line is determined based on voltage division between the source resistance R_s and the resistance seen by the signal looking into the T-line. However, unlike the steady-state case, the transient signal has no knowledge of how the T-line is terminated. It only sees R_o. So a voltage

$$V_o = V_s \frac{R_o}{R_o + R_s} \tag{2.100}$$

is initially launched on the line. Since the voltage in general will depend on the location along the line as well as time, it will be represented as $V(z,t)$. The initial launched voltage can therefore be written

$$V(-\ell, 0) = V_o \tag{2.101}$$

The *transit time* t_ℓ, or time for the signal to traverse the length ℓ of the line, is related to propagation velocity u_p by

$$t_\ell = \frac{\ell}{u_p} \tag{2.102}$$

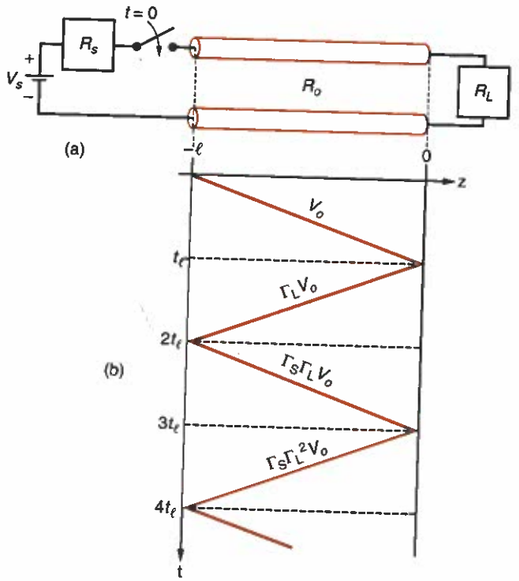


FIGURE 2.35 (a) A voltage step function is realized by closing the switch at $t = 0$. (b) Bounce diagram for transient voltage on the T-line.

Upon reaching the load end at time t_ℓ, part of the signal gets reflected. The reflection coefficient is

$$\Gamma_L = \frac{R_L - R_o}{R_L + R_o}$$

and a wave $\Gamma_L V_o$ is reflected. Therefore, the total voltage at the load end, just after the incident voltage has reached the load, is

$$V(0, t_\ell) = V_o(1 + \Gamma_L) \tag{2.103}$$

The reflected signal travels back toward the source. An additional t_ℓ time is required to reach the source end, where it sees a reflection coefficient

$$\Gamma_S = \frac{R_s - R_o}{R_s + R_o} \tag{2.104}$$

and a portion of $\Gamma_L V_o$ is reflected back toward the load. We therefore have

$$V(-\ell, 2t_\ell) = V_o(1 + \Gamma_L + \Gamma_L\Gamma_S) \tag{2.105}$$

With all the reflections and rereflections, keeping track of $V(z,t)$ can be difficult. To ease this difficulty we can employ a *bounce diagram*, also referred to as a reflection diagram, as shown in Figure 2.35(b). The position along the T-line is shown as the horizontal axis, here aligned for convenience with the T-line of Figure 2.35(a). Time is on the vertical axis. The diagonals represent the location of the leading edge of a wave. The topmost diagonal, traveling from time $t = 0$ to $t = t_\ell$, shows the location of the front of the incident wave V_o. The second diagonal shows the front of the first reflected wave $\Gamma_L V_o$, and so on. A vertical line drawn on the bounce diagram shows the voltage at that location as a function of time. At a particular time on the vertical line, the total voltage will consist of all the waves crossing the vertical above that point.

EXAMPLE 2.9

An example will show how a step change in voltage propagates along a section of T-line and how the bounce diagram works. We want to plot the voltage at the middle of the 6.0-cm-long T-line shown in Figure 2.36(a) as a function of time to 8.0 ns.

Our first step is to calculate the reflection coefficients. At the load, we have

$$\Gamma_L = \frac{125\Omega - 75\Omega}{125\Omega + 75\Omega} = \frac{1}{4}$$

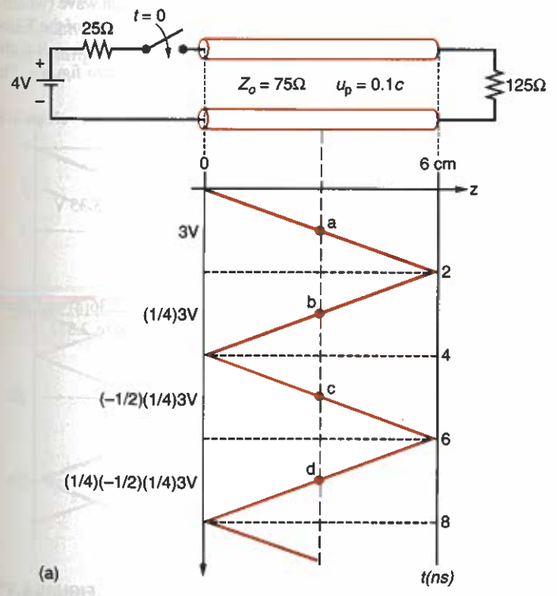


FIGURE 2.36a (a) T-line circuit and bounce diagram for a voltage step change.

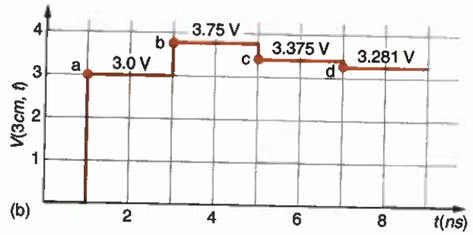


FIGURE 2.36b (b) Plot of the voltage at the middle of the T-line.

At the source end, we have

$$\Gamma_S = \frac{25\Omega - 75\Omega}{25\Omega + 75\Omega} = -\frac{1}{2}$$

We can also calculate the transit time for the 2.0-cm line, given a propagation velocity of 0.1*c*:

$$t_\ell = \frac{6\,\text{cm}}{(0.1)(3\times 10^8\,\text{m/s})}\left(\frac{1\,\text{m}}{100\,\text{cm}}\right)\left(\frac{10^9\,\text{ns}}{\text{s}}\right) = 2\,\text{ns}$$

Next, we indicate on the bounce diagram the values for each wave (where V_o=3V), shown to the left of the diagram in Figure 2.36(a). We tally these up at the middle of the T-line and plot the result in Figure 2.36(b). Initially, there is no voltage seen in the middle of the line. But after 1 ns has elapsed, the incident wave V_o is present, as indicated by point *a* on the two figures. This voltage remains constant until the reflected wave arrives, 2 ns later (point *b*).

After a long enough time, the reflections will settle down, and the voltage at every point along of the line will equal

$$V = V_s\frac{R_L}{R_L + R_s} = 4\,V\frac{125\,\Omega}{125\,\Omega + 25\,\Omega} = 3.33\,\text{V}$$

Drill 2.19 Swap the source and the load resistors for Figure 2.36(a) and generate a plot of the voltage at the source end of the line to 8 ns. (*Answer:* See Figure 2.37)

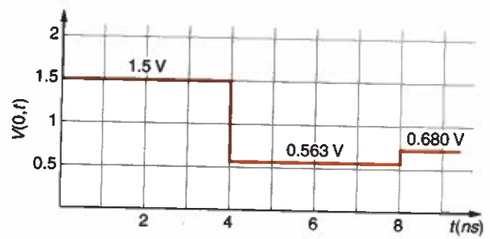


FIGURE 2.37 Plot of voltage at the source end for Drill 2.19.

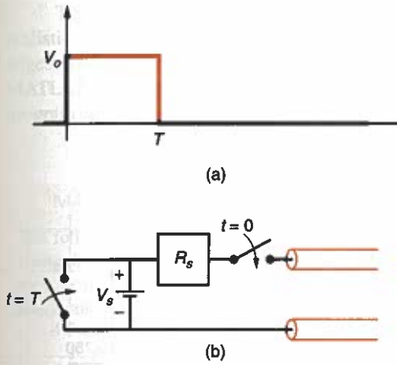


FIGURE 2.38 A pulse input as shown in (a) can be realized by the circuit in (b).

2.8.1 Pulse Response

Instead of a voltage step, suppose we launch the rectangular pulse shown in Figure 2.38(a) onto a T-line circuit. This can be represented by adding a second switch that shorts the supply at $t = T$, as shown in Figure 2.38(b).

We can model this situation by placing a pair of voltage step changes on the bounce diagram of Figure 2.39(a). A V_o incident wave is launched at $t = 0$, and it is turned off by launching a $-V_o$ wave at $t = T$. Finding the voltage at any point along the T-line at a

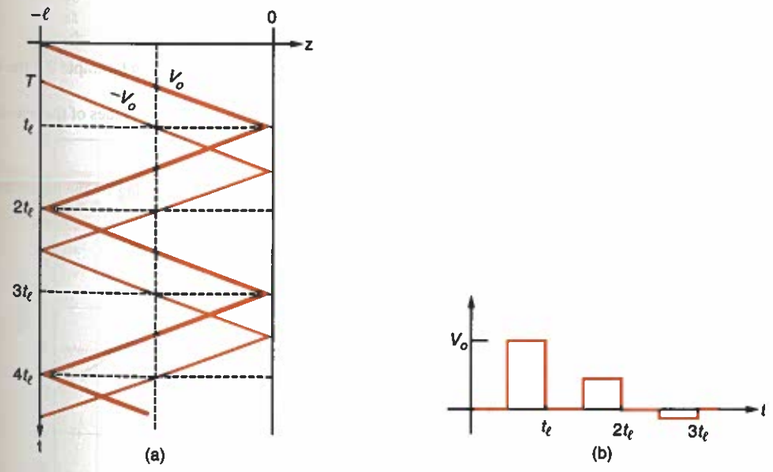


FIGURE 2.39 (a) Bounce diagram at the middle of the line for a pulse input. (b) Voltage at the middle of the line for a pulse input.

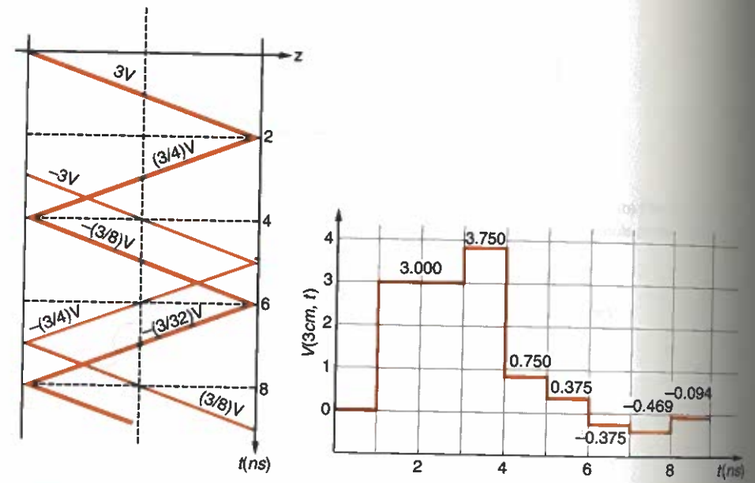


FIGURE 2.40 (a) The bounce diagram and (b) voltage plot for the circuit of Figure 2.36 (a) when a 4-V pulse of 3-ns duration is applied.

particular time follows the usual bounce-diagram approach. For instance, the hypothetical pulse response at the middle of the T-line is shown in Figure 2.39(b).

EXAMPLE 2.10

To better illustrate the pulse response procedure, suppose that in Example 2.9 the 4-V step voltage is replaced with a 4-V pulse of 3-ns duration.

The bounce diagram is shown in Figure 2.40(a) with the values of the waves as indicated. The voltage at the middle of the line is plotted in Figure 2.40(b). ■

Drill 2.20 For Example 2.10 with the 3-ns pulse, plot the voltage at the load out to 8 ns. (*Answer:* See Figure 2.41)

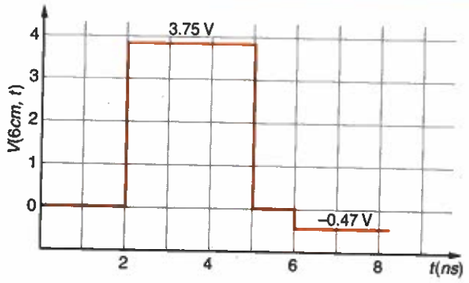


FIGURE 2.41 Plot for Drill 2.20.

The bounce diagram is a handy tool for step changes and rectangular pulses. However, realistic digital signals have significant rise and fall times, resulting in pulses with sloped edges. The bounce diagram becomes very difficult to use in such cases. The following MATLAB routine plots $V(z,t)$ resulting from a triangular pulse source. With modification, the program can be used for pulses with other shapes, including stepped and rectangular pulses.

MATLAB 2.5

The following routine traces the voltage at an arbitrary point along a T-line resulting from a triangular pulse. The pulse is to ramp up linearly from 0 to 5 V over 1.5 ns, then abruptly drop to zero, as indicated in Figure 2.42. We'll also consider the Example 2.9 situation of 6-cm line length and 0.1c propagation velocity. However, we'll short the load and want to find $v(z,t)$ at $z = 4.5$ cm.

To keep the routine simple, we choose to impedance match the source so that we only have a single positive traveling wave and, perhaps, a negative traveling wave. The program consists of two main sections. First, the triangular function is defined and placed into a time-discretized array. Second, the positive and the negative wave data are generated and combined.

```
%  M-File: ML0205
%  Analysis of a triangular pulse (matched source
%  impedance) traveling down a T-line and reflecting
%  off a resistive load. We want to be able to trace
%  the voltage at an arbitrary point along the line.
%
%  Wentworth, 4/25/03
%  Variables
%  Vo    pulse height (V)
%  t1    pulse start (ns)
%  t2    pulse end (ns)
```

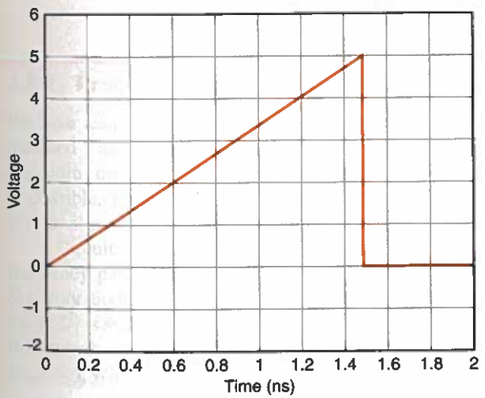


FIGURE 2.42 Triangular pulse source used in MATLAB 2.5.

```
%   L     line length (cm)
%   T     transit time (ns)
%   z     location to find pulse (cm)
%   tau   time "location" to find pulse (ns)
%   up    propagation velocity (m/s)

%   Ro,RL   line,load resistance (ohms)
%   N       number of points
%   GL      load reflection coefficient
clc
clear

%enter variables
Vo=5;
t1=0;
t2=1.5;
L=6;
z=4.5;
up=3e7;
Zo=50;
ZL=0;

T=1e9*(L/up)/100;
tau=1e9*(z/up)/100;

N=500;
GL=(RL-Ro)/(RL+Ro);
%initialize array
for i=1:N+1
  v(i)=0;
end

dt=2*T/N;
%enter triangular pulse function
m=0.5*Vo/(t2-t1); %slope
for i=1:N+1
  t(i)=i*dt;
  if t(i)<t1
    vo(i)=0;
  end
  if and(t(i)>t1,t(i)<t2)
    vo(i)=m*(t(i)-t1);
  end
  if t(i)>t2
    vo(i)=0;
  end
end

%Generate + wave data
for i=1:N+1
  ta=i*dt;
  if ta>tau
    j=ceil((ta-tau)/dt);
    vplus(i)=vo(j);
  end
end
```



```
%Generate - wave data
for i=1:N+1
  ta=i*dt;
  tb=2*T-tau;
  if ta>tb
    j=ceil((ta-tb)/dt);
    vmin(i)=GL*vo(j);
  end
end

%Sum the data
for i=1:N+1
  v(i)=vplus(i)+vmin(i);
end

plot(t,v)
xlabel('time (ns)')
ylabel('voltage')
AXIS([0 2*T -Vo Vo])
grid on
```

Note use of the function "ceil." Whereas the related function "round" will go to the nearest integer, the function "ceil" will round up, to keep $j > 0$.

Suppose we want to use the routine to model the pulse function of Example 2.10 (except with a matched source—See Problem 2.45). We create the unit step function:

```
function U=step(t,T)
%Unit step function. When time t exceeds a time T,
%value of the function is 1.
U=t>=T;
```

Then we modify the program, replacing the "enter triangular pulse function" with "enter rectangular pulse function":

```
%enter rectangular pulse function
t=0:dt:2*T;
vo=0.5*Vo*(step(t,t1)-step(t,t2));
```

2.8.2 Practical Application: Schottky-Diode Terminations

We have seen that signal reflection can be eliminated by terminating the end of the line in a matched load. But the load impedance in a digital circuit may be unknown, or it may vary depending on its logic state. Even if the load impedance is known, and a resistive termination is possible, for many digital circuits the presence of matching resistors will result in continual unwanted power consumption.

Circuit designers have used a number of termination strategies to improve high-frequency performance in digital interconnects. A popular technique employs a pair of Schottky diodes placed at the load end of a transmission line as shown in Figure 2.43. The diode D_1 is said to *clamp* to the supply voltage V_{cc} whereas D_2 clamps to ground. Consider the case where a pulse of value V_{cc} is incident from the transmission line onto an open circuit ($Z_L = \infty$). Without D_1, the reflected pulse would result in $2V_{cc}$ at the load end for the duration